

Overview of Natural language processing

Subject: Natural language processing

1.

Lexical semantics What is the computational meaning of individual words in context? **Distributional semantics** How can we learn semantic representations from data? **Named entity recognition (NER)** Given a stream of text, determine which items in the text map to proper names, such as people or places, and what the type of each such name is (e.g. person, location, organization). Although capitalization can aid in recognizing named entities in languages such as English, this information cannot aid in determining the type of named entity, and in any case, is often inaccurate or insufficient. For example, the first letter of a sentence is also capitalized, and named entities often span several words, only some of which are capitalized. Furthermore, many other languages in non-Western scripts (e.g. Chinese or Arabic) do not have any capitalization at all, and even languages with capitalization may not consistently use it to distinguish names. For example, German capitalizes all nouns, regardless of whether they are names, and French and Spanish do not capitalize names that serve as adjectives. This task is also referred to as token classification. **Sentiment analysis** (see also **Multimodal sentiment analysis**) Sentiment analysis involves identifying and classifying the emotional tone expressed in text. This technique involves analyzing text to determine whether the expressed sentiment is positive, negative, or neutral. Models for sentiment classification typically utilize inputs such as word n-grams, Term Frequency-Inverse Document Frequency (TF-IDF) features, hand-generated features, or employ deep learning models designed to recognize both long-term and short-term dependencies in text sequences. The applications of sentiment analysis are diverse, extending to tasks such as categorizing customer reviews on various online platforms. **Terminology extraction** The goal of terminology extraction is to automatically extract relevant terms from a given corpus. **Word-sense disambiguation (WSD)** Many words have more than one meaning; we have to select the meaning which makes the most sense in context. For this problem, we are typically given a list of words and associated word senses, e.g. from a dictionary or an online resource such as WordNet. **Entity linking** Many words—typically proper names—refer to named entities; here we have to select the entity (a famous individual, a location, a company, etc.) which is referred to in context.

2.

The premise of symbolic NLP is often illustrated using John Searle's Chinese room thought experiment: Given a collection of rules (e.g., a Chinese phrasebook, with questions and matching answers), the computer emulates natural language understanding (or other NLP tasks) by applying those rules to the data it confronts.

3.

$$RMM(token_N) = PMM(token_N) \times \frac{1}{2d} \left(\sum_{i=-d}^d ((PMM(token_N) \times PF(token_{N-i}, token_N, token_{N+i}))^i) \right)$$
$$\{RMM(token_{\{N\}})\} = \{PMM(token_{\{N\}})\} \times \frac{1}{2d} \left(\sum_{i=-d}^d ((PMM(token_{\{N\}}) \times PF(token_{\{N-i\}}, token_{\{N\}}, token_{\{N+i\}}))^i) \right)$$

4.

Natural language processing has its roots in the 1950s. Already in 1950, Alan Turing published an article titled "Computing Machinery and Intelligence," which proposed what is now called the Turing test as a criterion of intelligence, though at the time that was not articulated as a problem separate from artificial intelligence. The proposed test includes a task that involves the automated interpretation and generation of natural language.

5.

General tendencies and (possible) future directions Based on long-standing trends in the field, it is possible to extrapolate future directions of NLP. As of 2020, three trends among the topics of the long-standing series of CoNLL Shared Tasks can be observed:

6.

1950s: The Georgetown experiment in 1954 involved fully automatic translation of more than sixty Russian sentences into English. The authors claimed that within three or five years, machine translation would be a solved problem. However, real progress was much slower, and after the ALPAC report in 1966, which found that ten years of research had failed to fulfill the expectations, funding for machine translation was dramatically reduced. Little further research in machine translation was conducted in America (though some research continued elsewhere, such as Japan and Europe) until the late 1980s when the first statistical machine translation systems were developed. 1960s: Some notably successful natural language processing systems developed in the 1960s were SHRDLU, a natural language system working in restricted "blocks worlds" with restricted vocabularies, and ELIZA, a simulation of Rogerian psychotherapy, written by Joseph Weizenbaum between 1964 and 1966. Despite using minimal information about human thought or emotion, ELIZA was able to produce interactions that appeared human-like. When the "patient" exceeded the very small knowledge base, ELIZA might provide a generic response, for example, responding to "My head hurts" with "Why do you say your head hurts?". Ross Quillian's successful work on natural language was demonstrated with a vocabulary of only twenty words, because that was all that would fit in a computer memory at the time. 1970s: During the 1970s, many programmers began to write "conceptual ontologies", which structured real-world information into computer-understandable data. Examples are MARGIE (Schank, 1975), SAM (Cullingford, 1978), PAM (Wilensky, 1978), TaleSpin (Meehan, 1976), QUALM (Lehnert, 1977), Politics (Carbonell, 1979), and Plot Units (Lehnert 1981). During this time, the first chatterbots were written (e.g., PARRY). 1980s: The 1980s and early 1990s mark the heyday of symbolic methods in NLP. Focus areas of the time included research on rule-based parsing (e.g., the development of HPSG as a computational operationalization of generative grammar), morphology (e.g., two-level morphology), semantics (e.g., Lesk algorithm), reference (e.g., within Centering Theory) and other areas of natural language understanding (e.g., in the Rhetorical Structure Theory). Other lines of research were continued, e.g., the development of chatterbots with Racter and Jabberwacky. An important development (that eventually led to the statistical turn in the 1990s) was the rising importance of quantitative evaluation in this period.

7.

both statistical and neural network methods tend to focus more on the most common cases extracted from a corpus of texts, whereas the rule-based approach needs to provide rules for both rare and common cases equally. language models, produced by either statistical or neural networks methods, are more robust to both unfamiliar (e.g. containing words or structures that have not been seen before) and

erroneous input (e.g. with misspelled words or words accidentally omitted) in comparison to the rule-based systems, which are also more costly to produce. the larger such a (probabilistic) language model is, the more accurate it becomes, in contrast to rule-based systems that can gain accuracy only by increasing the amount and complexity of the rules leading to intractability problems. Rule-based systems are commonly used: